Real-time AI election monitoring for Armenia's 1,800 polling stations

Armenia is uniquely positioned to deploy the world's first nationwide AI-powered election monitoring system. Its Electoral Code already mandates fixed cameras at every polling station covering ballot boxes, entrances within 50 meters, and registration areas — infrastructure that most countries lack. A hybrid system combining computer vision with a lean human response team could monitor all 1,800 stations simultaneously at roughly one-fifth the cost of traditional human observers, with dramatically superior detection rates. This report evaluates global precedents, maps CV capabilities to Armenian electoral law, proposes a technical architecture, and models the economics of deployment for the upcoming 2026 Parliamentary Elections. coe

The core finding is sobering yet encouraging: no country has successfully deployed real-time AI video analytics at election scale. India's VMukti has equipped 65,000+ booths with webcasting, but automated anomaly detection remains limited. Armenia's existing camera mandate under Article 8.11.1 of its Electoral Code, combined with mature CV models like YOLO11 and person re-identification networks, makes a phased deployment technically feasible — though significant engineering, calibration, and legal groundwork remain. legislationline OSCE

Global precedents reveal a wide gap between webcasting and true AI monitoring

The global landscape of AI election monitoring divides into three maturity tiers: biometric voter verification (deployed), video webcasting with human monitors (widespread), and automated CV-based anomaly detection (largely aspirational).

India dominates the first two tiers. The Election Commission of India mandated 100% webcasting at all polling stations by June 2025, up from 50% in prior elections. VMukti Solutions, incubated at IIM-Ahmedabad, has deployed at **65,000+ booths** across India's 2019 Lok Sabha elections and claims 95%+ accuracy for people counting, 98%+ for facial recognition, and 94%+ for loitering detection. Their platform offers 26+ AI modules including crowd density mapping, weapons identification, and pre-stampede alerts, operating on a cloud-native VMS with edge AI cameras streaming at 20 kbps–4 Mbps via patented compression. However, these accuracy figures are self-reported and unverified by independent audits. In practice, India's webcasting system suffered 2–4 hour outages in West Bengal during the 2024 elections due to poor camera quality and rural connectivity. Business Standard + 6

Nigeria's biometric approach provides a contrasting model. The Bimodal Voter Accreditation System (BVAS) deployed across 177,000+ polling units uses dual biometric authentication (fingerprint plus facial recognition) to combat ghost voting. While effective in concept, widespread device failures, worn fingerprints, and poor lighting caused significant accreditation breakdowns in the 2023 elections. Iraq deployed 110,000+ CCTV systems from South Korea's MIRU SYSTEMS for its 2023 governorate elections, focused on physical security rather than fraud detection. Inecnigeria + 2

Academic research remains thin on real-time polling station CV. The most significant work is Cantú's 2019 CNN analysis of Mexico's 1988 tally sheets, achieving **89% overall accuracy** in detecting ballot alterations — but this was post-hoc forensic analysis, not real-time monitoring. An IEEE paper on face detection for elections reached ~90% accuracy for preventing multiple voting. Yamin et al. (2023) demonstrated agent-based election simulation with clustering for novelty detection. No published paper addresses real-time CV-based polling station surveillance — representing a genuine research gap and an opportunity for Armenia to lead. IDEAS/RePEc + 3

Commercial vendors like VIDIZMO market AI election monitoring (anomaly detection, crowd behavior analysis, tampering detection), but no confirmed large-scale election deployment of automated video analytics was found. Companies like BriefCam offer powerful forensic video analysis (video synopsis, appearance similarity, dwell time tracking) that could enhance post-election review but lack election-specific deployments. VIDIZMO MCA

System	Country	Scale	AI Capability	Maturity
VMukti	India	65,000+ booths	Crowd, face, weapons, tracking	Production (webcasting); AI analytics partially deployed
BVAS	Nigeria	177,000+ units	Biometric face + fingerprint	Production; high failure rates
MIRU CCTV	Iraq	110,000+ systems	Physical security monitoring	Production; no AI analytics
VIDIZMO	US/Pakistan	Marketing stage	Anomaly detection, crowd behavior	Marketed; no confirmed election deployment
Cantú CNN	Mexico (academic)	50,000+ tally images	Ballot alteration detection (89%)	Research only; post-hoc

Armenia's Electoral Code provides a ready-made legal framework for CV deployment

Armenia's Electoral Code, substantially reformed in 2016, contains one of the most comprehensive legal mandates for election video surveillance anywhere in the world. **Article 8.11.1 requires fixed cameras at every electoral precinct** during National Assembly elections, with fields of view covering the ballot box, polling station entrance within a 50-meter radius, voter registration, ballot allocation, and vote counting. Video must be webcast in real-time and archived for parties and observer organizations. This existing mandate provides both the physical camera infrastructure and the legal authorization for CV analysis of existing feeds. OSCE coe

The Code defines at least 14 violation categories that computer vision could plausibly detect. The highest-feasibility applications map directly to documented problems. **Article 6** establishes voting secrecy as both a right and a responsibility, with "control over the free expression of will" explicitly prohibited — detectable through pose estimation identifying individuals looking at others' ballots or through object detection of ballots displayed outside booths. **Article 67** limits voter assistance to one person of the voter's choice, excluding observers, media, and commission members — a violation that person re-identification (Re-ID) can detect by flagging the same individual repeatedly entering booth areas with different voters. OSCE observation of the 2023 Yerevan elections documented this pattern specifically. Parliament + 2

The most impactful CV applications target violations that OSCE consistently documents:

Carousel voting (same person voting multiple times) is directly addressable through person Re-ID using OSNet or TransReID across camera feeds at the same and different precincts. The mandated entrance cameras with 50-meter coverage create ideal conditions for capturing individuals entering and exiting.

Unauthorized crowding within the 50-meter perimeter (prohibited under Article 22.3) was documented in **26 cases across 21 polling stations** in the 2023 Yerevan elections alone. Crowd density estimation is among the most mature CV capabilities, with models like CSRNet achieving reliable real-time counting. Tribunal Electoral + 2

Ballot stuffing — detecting multiple ballot insertions — is feasible through activity recognition models monitoring the ballot box area, which cameras are legally required to cover. Abnormal hand-to-box frequency patterns can trigger alerts with high confidence.

Organized voter transport (buses or vans repeatedly stopping near stations) correlates with vote-buying schemes and is detectable through vehicle detection on entrance cameras. OSCE's 2017 report found large groups near stations in **30% of observed cases**, with direct intimidation in several instances. OSCE PA Council of Europe

Critical legal constraints shape system design. Cameras must never capture individual ballot marking (Article 6). Person Re-ID across precincts involves biometric data processing requiring compliance with Armenia's 2015 Data Protection Law (though enforcement is weak, with maximum fines of ~$1,300). Real-time facial recognition is not explicitly authorized under the Electoral Code and would require additional legal framework. Recent January 2026 amendments gave the CEC discretion to deny observer accreditation based on "political neutrality" — a provision that could affect NGO-deployed CV systems. coe + 3

Violation	Electoral Code Article	CV Detection Method	Feasibility
Carousel voting	Art. 10.2, Criminal Code	Person Re-ID (OSNet-AIN)	High
Ballot stuffing	Art. 67	Activity recognition (X3D-M) + tracking	High
Unauthorized crowding (50m zone)	Art. 22.3, Art. 19	Crowd density estimation	High
Booth occupancy violation	Art. 67 (one assistant only)	Person counting + tracking	High
Repeated voter assistance	Art. 67	Person Re-ID + tracking	High
Campaign materials on voting day	Art. 19.1	Object detection + OCR	Medium-High
Voter coercion/intimidation	Art. 4, Art. 6	Pose estimation + activity recognition	Medium
Ballot photographing	Art. 6	Object detection (phone in booth)	Medium
Ballot box tampering	Art. 64	Activity recognition + anomaly detection	Medium
Vote buying near station	Art. 19.6, Criminal Code	Object detection (hand exchanges)	Low-Medium

A hybrid edge-cloud architecture can process all 1,800 streams at 5–10 FPS

The technical architecture must balance three competing constraints: real-time latency (sub-3-second alerts), Armenia's variable rural connectivity, and cost efficiency for a 14-hour election day. A three-tier hybrid design achieves this.

Model selection optimized for surveillance conditions

The recommended inference pipeline chains five specialized models, each selected for performance on low-resolution, angled, ceiling-mounted cameras typical of polling stations:

YOLO11-m (Ultralytics) serves as the primary person and object detector, achieving ~51% mAP on COCO with **4.7ms inference** on a T4 GPU. Its native TensorRT and DeepStream integration, combined with the Ultralytics ecosystem for training and export, makes it the most deployment-ready option. For low-resolution 640×640 input from surveillance cameras, the accuracy gap versus heavier models like RT-DETR (53.1% mAP) narrows significantly.

BoT-SORT handles multi-object tracking by fusing Kalman filter motion prediction with appearance-based Re-ID features, achieving **HOTA 65 and IDF1 80 on MOT17** — critical for maintaining person identity through occlusions caused by furniture, ballot boxes, and other voters. ByteTrack serves as a lightweight fallback for lower-occupancy streams.

OSNet-AIN (Omni-Scale Network with Adaptive Instance Normalization) is the cornerstone for carousel voting detection. At just **2.2 million parameters** — 10× smaller than ResNet-based alternatives — it achieves 94.8% Rank-1 accuracy on Market-1501 with the best cross-domain generalization of any Re-ID model. This is essential since training data from Armenian polling stations will be limited. Its multi-scale feature learning focuses on body regions rather than faces, making it robust at surveillance resolution.

RTMPose-m provides selective pose estimation at **430+ FPS** on a GTX 1660Ti, enabling detection of suspicious body orientations (looking at others' ballots, aggressive postures) without becoming a processing bottleneck. **X3D-M** handles action recognition for flagged clips, achieving 76% Top-1 on Kinetics-400 at just 6.2 GFLOPs — 4.8× more efficient than SlowFast.
Readthedocs

Infrastructure requirements and cost

NVIDIA DeepStream 8.0 benchmarks show an L40S GPU processing **486 FPS** end-to-end for person detection plus tracking. Accounting for the multi-model pipeline overhead factor of ~1.65× and targeting 5–10 FPS inference per stream (sufficient for detecting ballot stuffing and crowd anomalies), each L40S handles **29–59 streams**.

Parameter	Conservative (15 FPS)	Balanced (10 FPS)	Efficient (5 FPS)
Streams per L40S GPU	19	29	59
Total L40S GPUs needed	119	78	39
Aggregate bandwidth	5.4 Gbps (1080p)	2.7 Gbps (720p)	2.7 Gbps (720p)
Storage (14 hrs, 2× redundancy)	70 TB	36 TB	36 TB
End-to-end alert latency	0.2–0.6s (urban)	0.5–2.6s (rural)	0.5–3s (rural)

The recommended architecture deploys **15 regional aggregation servers** (each with 2× L40S, handling ~120 stations) plus a **central GPU cluster of 8–12 servers** (4× L40S each) for the full multi-model pipeline. Total GPU count: **~50–78 L40S units**.

The recommended technical stack

The complete pipeline uses **NVIDIA DeepStream SDK 8.0** for hardware-accelerated video decode and batching, **TensorRT 10.x** for INT8/FP16 model optimization, **NVIDIA Triton Inference Server** for multi-model serving with dynamic batching, **Apache Kafka 3.7+** for event streaming and metadata, and **Redis Streams 7.x** for sub-10ms alert delivery to the dashboard. Kubernetes orchestrates GPU pod scheduling with auto-scaling. The dashboard runs on React with WebSocket connections for real-time operator interfaces. NVIDIA

Cloud cost for 14-hour election day compute alone runs $13,000–$18,000 in a pure cloud configuration (AWS g6.12xlarge instances at $4.60/hr). The hybrid configuration with rented regional servers reaches $60,000–$80,000 but provides critical resilience against connectivity failures in rural Armenia. For context, AWS cut A100 instance prices by 33–45% in June 2025, making GPU compute increasingly accessible. Data Center Dynamics AWS

The most significant technical risk is **network connectivity in rural Armenia**, where many stations rely on 4G/LTE with 5–20 Mbps variable bandwidth. Streams should be transcoded to 720p at 15 FPS (1.5 Mbps per stream) at the source. Local recording via ring buffers on station routers provides a failsafe, and adaptive bitrate encoding can reduce streams to sub-1 Mbps in constrained conditions — a technique VMukti has already proven at 20 kbps minimum in Indian elections. VIDIZMO Scribd

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The hybrid model cuts observer costs by 80% while improving detection quality

The economics of AI-augmented election monitoring are compelling, but the operational case is even stronger. Research on vigilance decrement shows that **human CCTV operators overlook ~45% of activity after 12 minutes and up to 95% within 22 minutes**. Over a 14-hour election day, even with three observers per station rotating through shifts, attention quality degrades severely — and the evening counting period, when fraud is most likely, coincides with peak fatigue. ScienceDirect

Cost comparison across three models

A pure human observer deployment covering all 1,800 stations requires approximately **5,400 personnel** (3 per station for rotation) at an estimated $40/day in Armenian wages, plus training, transportation, and coordination — totaling **~$537,000 per election**. The AI system's development cost of $590,000–$930,000 is front-loaded, but per-election operational costs drop to just $8,000–$18,000 for compute, storage, and bandwidth.

The recommended hybrid model deploys AI monitoring across all 1,800 stations with a **25–30 person command center** in Yerevan, **200 mobile field responders** organized into 40 geographic teams, and **500 static observers** at historically high-risk stations. Total personnel: ~750, at a per-election cost of **~$102,000–$112,000**.

Metric	Human-Only	AI-Only	Hybrid (Recommended)
Per-election cost	~$537,000	~$8,000–$18,000	**~$107,000**
Year 1 (incl. development)	~$537,000	$678K–$1.46M	**$770K–$1.14M**
10-year / 5-election total	~$2.68M	$1.06–$2.27M	**$1.26–$1.76M**
Station coverage	85–90%	100%	**100%**
Detection rate (trained patterns)	~20% (research baseline)	90–95%	**85–95%**
Novel violation detection	Good	Poor	**Good**
Legal evidentiary value	High	Uncertain	**High**

Break-even occurs after just two elections. Cumulative savings over five elections reach $900,000–$1.4 million compared to the human-only model.

Operational workflow and false positive management

The system operates through a four-tier alert protocol. **Green alerts** (normal operations) are logged silently. **Yellow alerts** (60–80% anomaly confidence: unusual clustering, extended ballot box time) queue for analyst review within 5–10 minutes. **Orange alerts** (80–95% confidence: potential ballot stuffing patterns, camera tampering) trigger immediate analyst review within 2 minutes. **Red alerts** (>95% confidence or multiple corroborating signals: confirmed stuffing patterns, physical altercations, camera destruction) escalate instantly with field team dispatch.

Modern AI video analytics achieve **85–95% false alarm reduction** over traditional motion detection. At a realistic 95–98% precision rate, the system would generate approximately **25–100 false positives across the entire 14-hour election day** — translating to 0.5–4 false alarms per analyst, a manageable volume. Pre-election calibration using 2–3 days of mock scenarios, multi-signal correlation requirements (demanding 2+ independent indicators before escalation), and station-specific baseline learning further reduce operational false positive impact. Scylla +3

Command center analysts work **2-hour active shifts** with 30-minute breaks, aligned with vigilance research showing significant attention decay beyond this threshold. Each primary analyst monitors AI alerts for ~120 stations — a manageable load given that the AI pre-filters routine activity and only surfaces anomalies. ScienceDirect

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Feasibility assessment and recommended phased deployment

The system is **technically feasible but operationally ambitious** for a first deployment. Three factors favor Armenia specifically: the existing legal mandate for comprehensive camera coverage, the manageable scale of 1,800 stations (versus India's million-plus), and documented recurring violations that CV can reliably detect.

Phase 1 (2026 election): Deploy AI monitoring as a **parallel shadow system** at 200 high-risk stations alongside full human observer coverage. Validate detection rates against human observations. Calibrate false positive thresholds. Collect training data from Armenian polling station footage. Estimated cost: $250,000–$400,000 (development plus pilot infrastructure).

Phase 2 (next election cycle): Scale AI to all 1,800 stations. Reduce static observers to 500 high-risk stations plus 200 mobile responders. Operate the full hybrid model with the central command center. Estimated per-election cost: ~$107,000.

Phase 3 (subsequent elections): Continuous model improvement with accumulated Armenian training data. Expand to pre-election security monitoring (equipment tampering, station setup compliance). Explore cross-precinct Re-ID for carousel voting detection across the entire network. Add edge preprocessing at 300 priority stations with Jetson Orin Nano devices. NVIDIA

Critical success factors

Five elements determine whether this system delivers on its promise:

- **Camera quality and placement standardization** across all 1,800 stations — India's experience shows that unreliable cameras and poor angles undermine even basic webcasting, let alone AI analytics New Kerala
- **Rural connectivity resilience** — adaptive bitrate encoding, local recording buffers, and regional aggregation servers are non-negotiable for Armenia's geography
- **Domain-specific fine-tuning** — general-purpose models must be calibrated on Armenian polling station footage; even a small annotated dataset from prior elections dramatically improves performance
- **Legal and political buy-in** — the CEC must authorize CV analysis of mandated camera feeds, and data protection compliance must be established before deployment
- **Human-in-the-loop design** — AI generates alerts, humans make determinations; this is both an operational necessity (for handling edge cases) and a legal requirement (for evidentiary admissibility) Securitysystemsauthority Omnilert

Conclusion

Armenia's combination of mandated polling station cameras, documented recurring violations detectable by CV, and a manageable 1,800-station scale creates a genuinely viable opportunity for the world's first nationwide AI election monitoring system. The technology stack — YOLO11 detection, BoT-SORT tracking, OSNet-AIN re-identification, RTMPose pose estimation, and X3D action recognition running on NVIDIA DeepStream — is mature enough for production deployment at this scale with 40–80 L40S GPUs. The hybrid model's **80% cost reduction** versus human observers, combined with 100% station coverage and sub-3-second alert latency, represents a step change in election integrity capability. The system's greatest limitation is not technical but institutional: success depends on political will to fund development, standardize camera infrastructure, and establish the legal framework for CV-generated evidence. A phased pilot starting with 200 stations in 2026 would validate the approach with limited risk while positioning Armenia as a global leader in technology-assisted electoral transparency.